Hardness test — data sheet

UMT 312L · Group 1 (Friday), handout **Gp 4** · Mayank Bohat, Chidambar · 13 November 2026

$HV = 1.8544\,F/d^2$ $HB = 2F/\pi D\left(D - \sqrt{D^2 - d^2}\right)$, $D = 2.5$ mm F in kgf, d in mm · —— = not measured

1. Material type (Brinell and Vickers)

MS, Cu, SS · Brinell 2.5 mm ball at 187.5 kgf and Vickers at 100 kgf

Mild steel

	Brinell, 187.5 kgf		**Vickers**, 100 kgf	
indent	d_1 (mm)	d_2 (mm)	d_1 (mm)	d_2 (mm)
1	1.050	1.030	0.923	0.922
2	1.027	1.038	0.918	0.913§
3	1.044	1.021	0.922	0.924
mean d	**1.0350**		**0.9203**	
hardness	**212.9** HB		**218.9** HV	
95 %	± 4.6		± 5.0	

Stainless steel

	Brinell, 187.5 kgf		**Vickers**, 100 kgf	
indent	d_1 (mm)	d_2 (mm)	d_1 (mm)	d_2 (mm)
1	0.873	0.869	0.799	0.771
2	0.879	0.882	0.776	0.766
3	0.853	0.862	0.791	0.770
mean d	**0.8697**		**0.7788**	
hardness	**305.8** HB		**305.7** HV	
95 %	± 20.9		± 14.0	

Copper

	Brinell, 187.5 kgf		**Vickers**, 100 kgf	
indent	d_1 (mm)	d_2 (mm)	d_1 (mm)	d_2 (mm)
1	1.474	1.466	1.377	1.328
2	1.459	1.470	1.352	1.349
3	1.462	1.480	1.386	1.337
mean d	**1.4685**		**1.3548**	
hardness	**100.1** HB		**101.0** HV	
95 %	± 1.3		± 2.2	

2. Effect of load (MS, Brinell)

mild steel, Brinell 2.5 mm ball · three indents at each load

Load (kgf)	**21.2**	**60**	**100**	**150**	**187.5**
indent 1, d_1/d_2 (mm)	0.415/0.442	0.601/0.608	0.754/0.760	0.939/0.929	1.039/1.038
indent 2, d_1/d_2 (mm)	0.444/0.443	0.588/0.606	0.797/0.768	0.931/0.946	1.035/1.031
indent 3, d_1/d_2 (mm)	0.448/0.447	0.600/0.604	0.729/0.730	0.935/0.933	1.042/1.035
mean d (mm)	**0.4398**	**0.6012**	**0.7563**	**0.9355**	**1.0367**
projected area $\pi d^2/4$ (mm^2)	0.1519	0.2838	0.4493	0.6873	0.8440
HB	**138.4**	**208.3**	**217.4**	**210.3**	**212.1**

3. Effect of heat treatment (Al–Cu, Vickers)

Vickers at 100 kgf

As-solutionized, 550 °C 1 h, water quenched

indent	d_1 (mm)	d_2 (mm)
1	1.176	1.236
2	1.234	1.187
3	1.245	1.178
4	1.249	1.202
5	1.247	1.189
6	1.234	1.189
mean d	**1.2138**	
hardness	**125.9** HV	
95 %	± 1.5	

Aged 230 °C — block A[‡]

indent	d_1 (mm)	d_2 (mm)
1	1.129	1.138
2	1.158	1.160
3	1.149	1.130
4	1.175	1.162
5	1.140	1.195
mean d	**1.1536**	
hardness	**139.3** HV	
95 %	± 4.9	

Aged 230 °C — block B[‡]

indent	d_1 (mm)	d_2 (mm)
1	1.153	1.143
2	1.132	1.136
3	1.149	1.141
4	1.143	1.166
5	1.138	1.165
mean d	**1.1466**	
hardness	**141.1** HV	
95 %	± 2.4	

*As-solutionized: 550 °C for 1 hr, then water quenched.